Arrival-delay labels at the minute boundary

Independent research program 31 October 2026 UTC

Not peer reviewed. One current source view; fixed recorded-minute probes.

Practical finding

Changing a delay cutoff changes the target even when the flights stay the same. In the fixed January 2024 reporting-carrier cohort, 275 of 44,707 eligible records change labels between recorded cutoffs of 15 and 16 minutes.

At ORD the affected count is 131 of 18,805, a 0.697 percentage-point rate difference. At ATL it is 144 of 25,902, or 0.556 percentage points. These are sensitivity results on recorded minutes, not a reconstruction of a European rate or physical-second lateness.

Recommended use

Keep the source field, numeric cutoff and eligible denominator with a benchmark label. Rebuild the label from supported source observations before pooling systems. Retain cancellation, diversion and unknown-field accounts alongside completed-flight rates. A matching label name or published aggregate is insufficient evidence that the target and population match.

The local implementation returns qualified recorded-minute counts and rejects unsupported requests to infer physical seconds, European OTP15, safety rankings or a transfer score. This is a runnable implementation exercise, not a deployed forecasting service or measured operating benefit.

Source and fixed scope

The official BTS January 2024 reporting-carrier archive contains 547,271 national rows. Every arrival to the two fixed destination codes is retained: 46,621 records in 21,303 day/carrier/source-state count cells. Native archives and flight identities remain private.

The public protocol fixed month, destinations, eligibility and cutoffs before flight-record retrieval. After the source form failed, the official archive directory supplied a published direct link to the identical fixed file. That transport change was committed before reading its body. The earlier failure is preserved; no month, airport or reporting-population substitution was made.

Original transfer question remains unexecuted

The original US-to-Europe model-transfer study remains unexecuted. Common-event, reporting-population, source-rights and historical-vintage requirements remain unresolved. The 16-minute probe is an authored comparison on US recorded values. It does not equate US minutes with European physical-second labels. No European source values are transformed or scored.

Complete airport accounts and denominators

The rate denominator contains eligible completed nondiverted source records. Negative early arrivals remain eligible. The early-arrival-clamped field was not substituted. Missing or malformed values remain explicit.

Source account	ORD	ATL
All selected source records	20,327	26,294
Eligible completed nondiverted	18,805	25,902
Missing key	0	0
Duplicate key	0	0
Unknown operation flags	0	0
Both operation flags set	0	0
Cancelled	1,479	311
Diverted	43	81
Unknown delay	0	0
Usable official indicator among eligible	18,805	25,902
Unknown official indicator among eligible	0	0
Official indicator mismatch among eligible	0	0

All fixed recorded-minute thresholds

Cutoff	Quantity	ORD	ATL
Recorded delay >= 14	Positive count	5,122	5,571
	Percent of eligible	27.24%	21.51%
Recorded delay >= 15	Positive count	4,989	5,377
	Percent of eligible	26.53%	20.76%
Recorded delay >= 16	Positive count	4,858	5,233
	Percent of eligible	25.83%	20.20%

Fifteen follows the source indicator definition; 14 and 16 are prespecified one-minute probes, not optimized replacements. The primary rate difference is the exact recorded bin $15 \leq \text{ArrDelay} < 16$ divided by the eligible count. As a share of the recorded >=15 positives, that bin is 2.63% at ORD and 2.68% at ATL. A zero denominator is unavailable, never silently zero.

Complete case retention

The repository exports all 62 airport-day accounts, 26 airport/carrier accounts and 270 threshold cases. Every day is retained. The carrier grid includes zero-row combinations with unavailable rates and all source code labels; it does not rank carrier quality. Agreement with the source indicator in this finite cohort does not establish general source accuracy, physical seconds or cross-system equivalence.

Verification, implementation and next evidence

Independent evidence

A separate pandas C parser, SQLite key classification and exact Fraction values reconcile the captured native source in 131,867 finite conditions. Saved native-bound source checks pass 191,424. An independent weighted SQLite/Fraction checker reconstructs all accounts, thresholds, numerators, denominators and example requests in 2,947 conditions without importing the production metric functions.

The committed freeze binds 26 actual source/method dependencies before threshold comparisons. Source values were visible during capture and reconciliation; this was source informed, not blinded. The 29 authored fixture methods cover exact cutoff and fractional-minute values, negative delay, unknown fields, operation flags, duplicate groups, zero denominators, indicator disagreement and unsupported interpretations.

Separate full-history CPython 3.10.11 and 3.12.14 replays reproduce 6 result files exactly, preserving 10 source-directory files in each. Figure checks bind 90 bars and 360 actual SVG coordinate conditions. These finite checks are not universal correctness guarantees.

Runnable implementation exercise

Seven complete requests accept two qualified recorded-minute rate descriptions and reject five unsupported requests. The interface retains the actual numerator, eligible denominator and exact ratio, plus explicit unknown physical-second and European equivalence. It checks supplied verified accounts; it does not certify arbitrary caller input.

```
python -m pip install -r requirements.txt
python src/replay.py --out ../arrival-label-replay
```

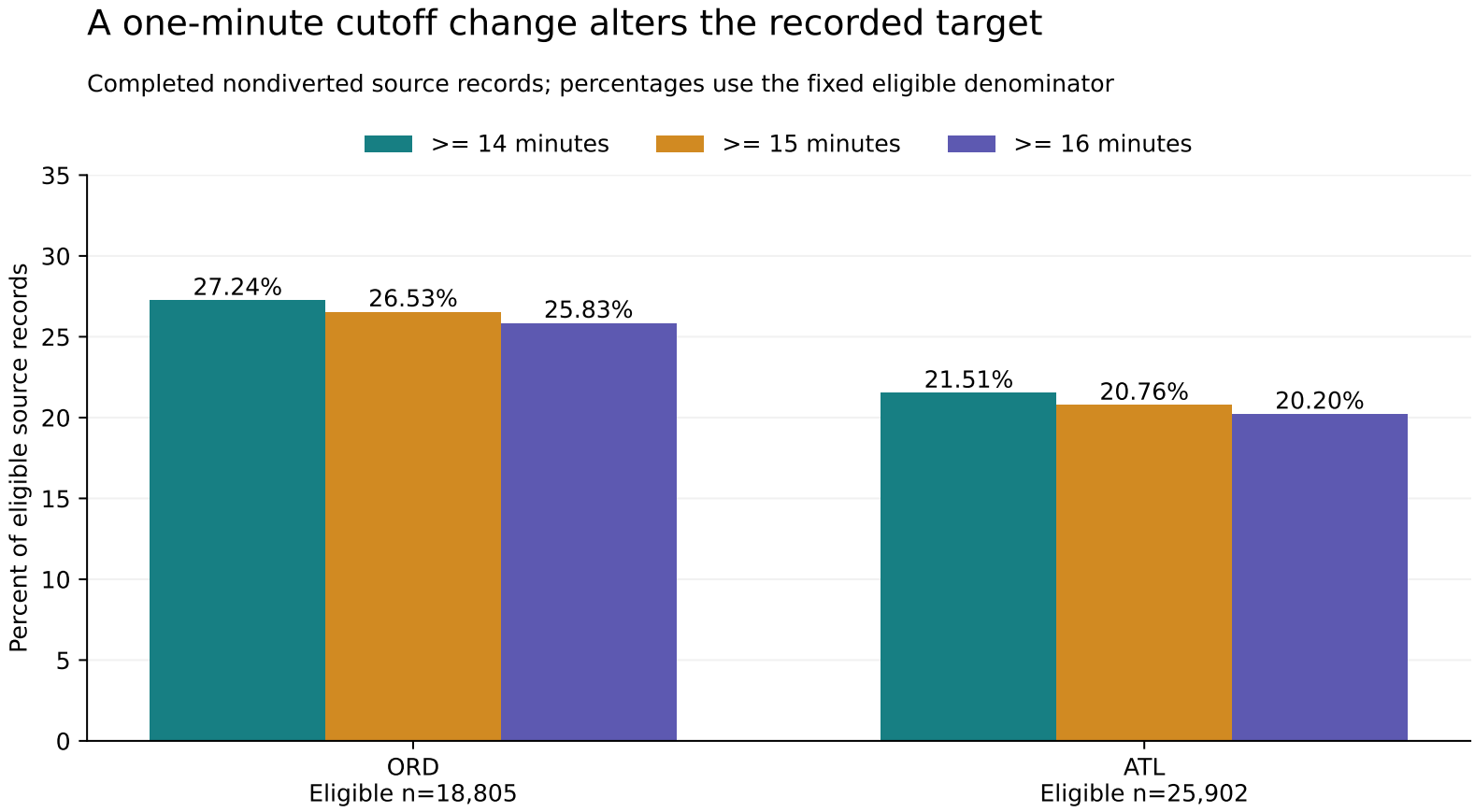
Use a new output directory. The replay checks frozen saved source counts; it does not reacquire the archive or establish original publication timing.

Material limits

One winter month, two destination codes and one reporting-carrier source form a convenience cohort. Current historical data can be revised, and the dataset is not an immutable original-vintage capture. Recorded-minute values do not recover the physical-second distribution. No cross-region calibration, causal airport effect, carrier safety/quality score, new estimator, production deployment or realized saving is claimed.

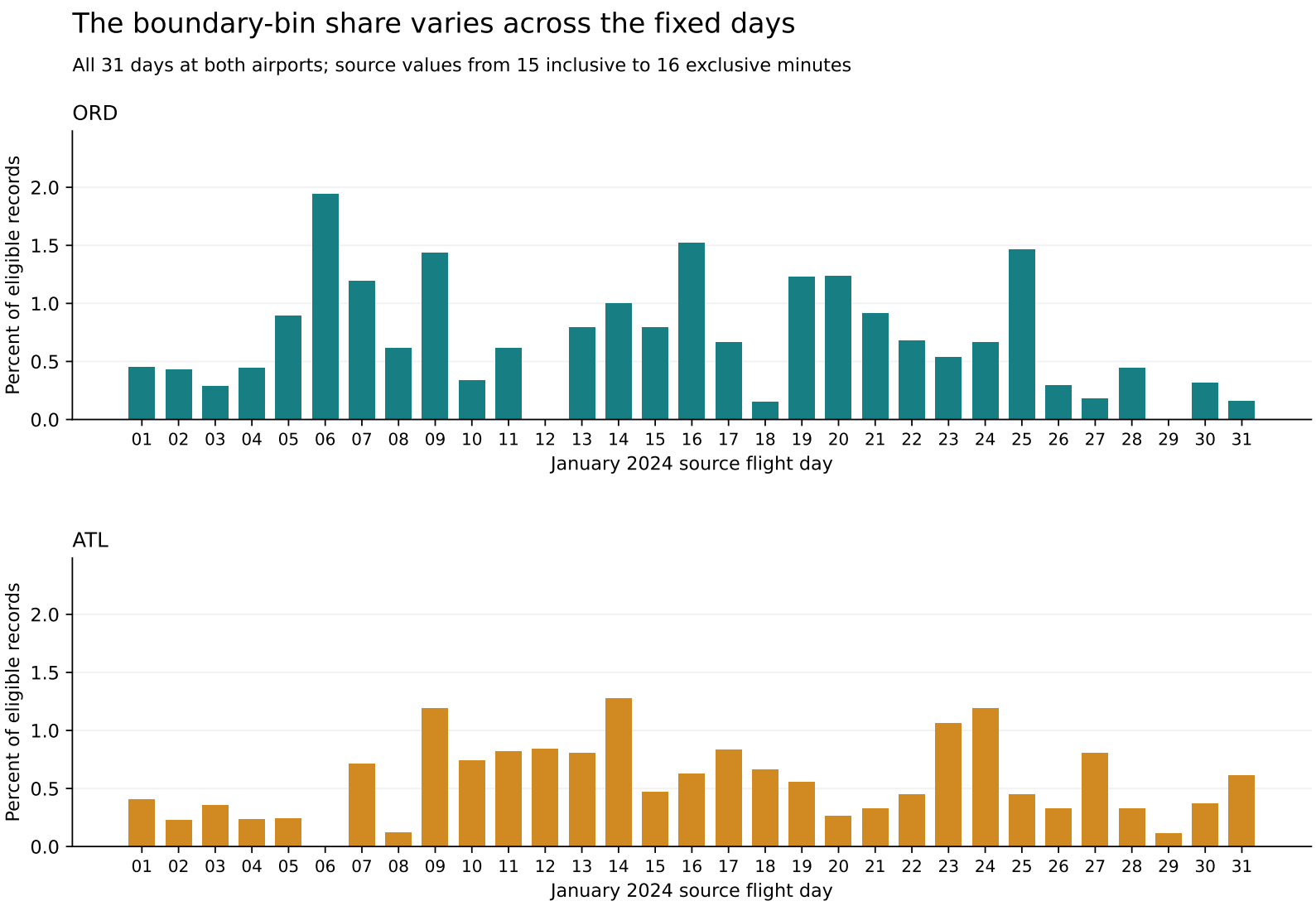
A transfer follow-up needs threshold-resolved compatible targets, aligned operated/cancelled populations, suitable source rights and verified dated availability. Formal-publication projects, including Azure optimization, remain private. Public artifacts contain minimal federal statistical counts and original analysis; native flight identities, tail numbers and source files remain private.

Source: U.S. Department of Transportation, Bureau of Transportation Statistics. https://transtats.bts.gov/PREZIP/. Official field definitions, exact indexed archive URL, capture/member hashes and failed-form history are retained in the repository. This independent study is not endorsed by DOT/BTS. MIT covers original code; source facts are attributed separately.



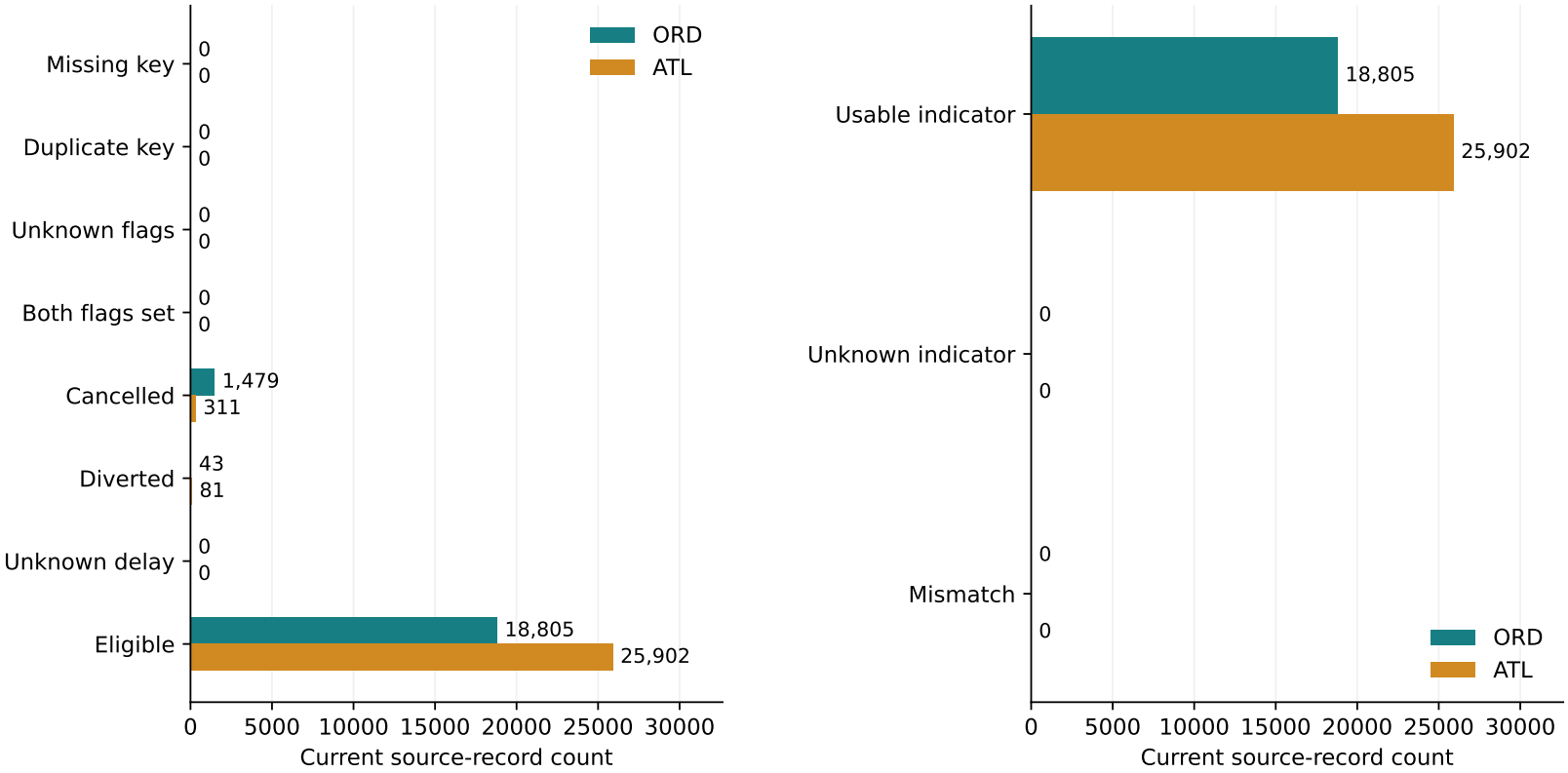
One January 2024 source view. These probes do not identify physical seconds or European OTP15.

Every fixed source day



Denominators retain cancellations, diversions and unknowns

Left: complete source eligibility partition. Right: eligible-record indicator diagnostics, which can overlap.



Zero categories remain visible as numeric labels. Source-indicator agreement is not proof of physical-second timing.